

General- m Minimal-Equilibrium Stability

Explicit sufficient regions, the bifurcation boundary, and an exact $m = 3$ counterexample

27 July 2026

Abstract

We study the one-dimensional Neumann minimal chemotaxis system with mobility u^m . The sufficient stability conditions are written explicitly. For $1 \leq m < 2$, two energy routes give global attraction on the conserved-mass hyperplane. At $m = 2$, the same conclusion holds under a sharp parameter-level endpoint admissibility condition. For every $m > 1$, a negative-power oscillation estimate gives a separate small-sensitivity theorem.

The boundary $m = 2$ belongs to the energy method, not to the dynamics. A weakly nonlinear calculation determines the direction of the first Neumann steady branch. For

$$m = 3, \quad \beta = \gamma = \mu = \nu = U = 1,$$

a Banach-space implicit-function argument constructs a positive nonconstant steady state at a sensitivity strictly below the linear threshold. This steady state is an exact obstruction to global attraction. The proof is analytic and does not rely on numerical continuation.

Contents

1	Model and meaning of stability	2
2	Linear stability threshold	2
3	Explicit sufficient stability conditions	3
3.1	Constants appearing in the two energy routes	3
3.2	Proof of the entropy branch	4
3.3	Proof of the signal-energy branch	5
3.4	Why the same proof reaches $m = 2$	6
4	Every $m > 1$: a separate small-sensitivity route	6
5	Steady-state bifurcation	7
5.1	Integrated steady equation	7
5.2	Weakly nonlinear direction coefficient	8
6	An exact subcritical steady state at $m = 3$	8
6.1	Banach-space branch construction	8
6.2	Exact branch direction	9
6.3	Positivity and the dynamical obstruction	10
7	Resolved region and remaining gap	10

1 Model and meaning of stability

On $I = (0, 1)$, consider

$$u_t = u_{xx} - \chi_0 \partial_x \left(\frac{u^m}{(1+v)^\beta} v_x \right), \quad (1)$$

$$0 = v_{xx} - \mu v + \nu u^\gamma, \quad (2)$$

$$u_x(t, 0) = u_x(t, 1) = v_x(t, 0) = v_x(t, 1) = 0. \quad (3)$$

All parameters satisfy

$$m > 0, \quad \beta \geq 1, \quad \gamma, \mu, \nu > 0, \quad \chi_0 > 0.$$

Integration of (1) gives the conserved mass

$$\int_0^1 u(t, x) dx = U > 0.$$

The constant equilibrium on this mass hyperplane is

$$u_* = U, \quad v_* = \frac{\nu}{\mu} U^\gamma. \quad (4)$$

Here “global attraction” means the following statement about every positive, bounded, global classical solution with mass U :

$$\|u(t) - U\|_{L^\infty} \longrightarrow 0,$$

and there are orbit-dependent constants $C, \rho, t_0 > 0$ such that

$$\|u(t) - U\|_{C^1} + \|v(t) - v_*\|_{C^1} \leq C e^{-\rho t} \quad (t \geq t_0). \quad (5)$$

The positive entry time is necessary when the initial datum is only continuous.

2 Linear stability threshold

Let $\lambda_n = n^2 \pi^2$ be the Neumann eigenvalues. Write

$$u = U + p, \quad v = v_* + q.$$

For the n -th nonconstant mode, (2) gives

$$\hat{q}_n = \frac{\nu \gamma U^{\gamma-1}}{\mu + \lambda_n} \hat{p}_n.$$

Linearization of (1) therefore gives the growth rate

$$\sigma_n(\chi_0) = -\lambda_n + \chi_0 \frac{\nu \gamma U^{m+\gamma-1} \lambda_n}{(1+v_*)^\beta (\mu + \lambda_n)}. \quad (6)$$

Proposition 1 (linear threshold). *The constant equilibrium is linearly stable on the mass-zero subspace if*

$$\chi_0 < \chi_{\text{lin}} := \frac{(1+v_*)^\beta (\mu + \pi^2)}{\nu \gamma U^{m+\gamma-1}}. \quad (7)$$

At equality the first cosine mode is neutral, and it is the first nonconstant mode to lose stability.

Proof. For $n \geq 1$, $\sigma_n < 0$ is equivalent to

$$\chi_0 < \frac{(1 + v_*)^\beta (\mu + \lambda_n)}{\nu \gamma U^{m+\gamma-1}}.$$

The right-hand side is strictly increasing with λ_n , so its minimum occurs at $n = 1$. $\square$

For the normalized tuple $U = \beta = \gamma = \mu = \nu = 1$,

$$\chi_{\text{lin}} = 2(1 + \pi^2) \approx 21.73920880. \quad (8)$$

3 Explicit sufficient stability conditions

3.1 Constants appearing in the two energy routes

Define the sharp signal saturation factor

$$\sigma_\beta = \sup_{z \geq 0} \frac{z}{(1+z)^\beta} = \begin{cases} 1, & \beta = 1, \\ \frac{(\beta-1)^{\beta-1}}{\beta^\beta}, & \beta > 1, \end{cases} \quad (9)$$

and

$$\chi_\beta = \frac{2(2\beta-1)}{\max\{2, \gamma\}}. \quad (10)$$

The formula for σ_β follows by differentiating $z(1+z)^{-\beta}$; for $\beta > 1$ its unique maximum is $z = (\beta-1)^{-1}$.

The energy estimates use an eventual box. Namely, choose $\bar{u} > 0$ and $\underline{v} > 0$, depending only on the parameters and U , such that every orbit under consideration satisfies

$$u(t, x) \leq \bar{u}, \quad v(t, x) \geq \underline{v} \quad \text{for all sufficiently large } t, \text{ uniformly in } x. \quad (11)$$

For $1 \leq m < 2$, standard one-dimensional L^P recurrence, Agmon interpolation, the conserved mass, and positivity of the Neumann elliptic Green function produce such orbit-independent constants. The same construction works at $m = 2$ under the endpoint condition

$$\chi_0 \sqrt{\mu} \sigma_\beta U \max\{2, \gamma\} < 1. \quad (12)$$

The role of (12) is to leave room for a restarted exponent strictly larger than 2 and at least γ .

On the interval $[0, \bar{u}]$, define the power slope

$$L_\gamma(U, \bar{u}) = \begin{cases} U^{\gamma-1}, & 0 < \gamma \leq 1, \\ \gamma \bar{u}^{\gamma-1}, & \gamma > 1. \end{cases} \quad (13)$$

Then

$$|s^\gamma - U^\gamma| \leq L_\gamma(U, \bar{u}) |s - U|, \quad 0 \leq s \leq \bar{u}.$$

The entropy threshold is

$$\chi_E = \frac{2\sqrt{\mu} (1 + \underline{v})^\beta \bar{u}^{-m}}{\nu L_\gamma(U, \bar{u})}, \quad (14)$$

and the signal-energy threshold is

$$\chi_S = \min \left\{ \frac{\chi_\beta}{2}, \sqrt{\chi_\beta}, \frac{\mu(1 + \underline{v})^\beta}{\nu \bar{u}^m} \right\}. \quad (15)$$

Theorem 1 (two explicit stability branches). *Assume $1 \leq m < 2$, and let $\bar{u}, \underline{v}$ satisfy (11). The equilibrium (4) is globally attracting if either*

$$0 < \chi_0 < \min\{\chi_{\text{lin}}, \chi_E\}, \quad (\text{E})$$

or

$$\gamma = 1, \quad 0 < \chi_0 < \min\{\chi_{\text{lin}}, \chi_S\}. \quad (\text{S})$$

At $m = 2$, the same conclusion holds under (12) and either (E) or (S).

3.2 Proof of the entropy branch

Define

$$H_m(s) = \int_U^s \left[1 - \left(\frac{U}{\tau} \right)^{2m-1} \right] d\tau, \quad \mathcal{E}_u(t) = \int_0^1 H_m(u(t, x)) dx. \quad (16)$$

For $m \geq 1$, $H_m \geq 0$ and $H_m(s) = 0$ only at $s = U$. Put

$$c_m = (2m-1)U^{2m-1}, \quad G = \int_0^1 u^{-2m} u_x^2, \quad X = \int_0^1 \frac{u^{-m} u_x v_x}{(1+v)^\beta}.$$

Differentiation of (16), followed by integration by parts, gives

$$\mathcal{E}'_u(t) \leq -c_m G + \chi_0 c_m X. \quad (17)$$

The boundary terms vanish because of (3).

Young's inequality, with

$$A = u^{-m} u_x, \quad B = (1+v)^{-\beta} v_x,$$

gives

$$-c_m G + \chi_0 c_m X \leq -\frac{c_m}{2} G + \frac{\chi_0^2 c_m}{2} \int_0^1 \frac{v_x^2}{(1+v)^{2\beta}}. \quad (18)$$

Inside the eventual box, mass conservation and the Neumann Poincaré inequality imply

$$G \geq \bar{u}^{-2m} \int_0^1 (u - U)^2. \quad (19)$$

The elliptic equation, tested against $v - v_*$, and the sharp one-dimensional multiplier bound give

$$\int_0^1 \frac{v_x^2}{(1+v)^{2\beta}} \leq \frac{\nu^2}{4\mu(1+\underline{v})^{2\beta}} \int_0^1 (u^\gamma - U^\gamma)^2. \quad (20)$$

Using (13),

$$\int_0^1 (u^\gamma - U^\gamma)^2 \leq L_\gamma(U, \bar{u})^2 \int_0^1 (u - U)^2.$$

Consequently,

$$\mathcal{E}'_u(t) \leq -C_E \int_0^1 (u - U)^2, \quad (21)$$

where

$$C_E = \frac{c_m}{2} \left[\bar{u}^{-2m} - \frac{\chi_0^2 \nu^2 L_\gamma(U, \bar{u})^2}{4\mu(1+\underline{v})^{2\beta}} \right]. \quad (22)$$

Condition $\chi_0 < \chi_E$ is exactly $C_E > 0$.

Since $\mathcal{E}_u \geq 0$, integration of (21) gives

$$\int_T^\infty \|u(t) - U\|_{L^2}^2 dt < \infty.$$

Thus there are arbitrarily late times t_j with $\|u(t_j) - U\|_{L^2} \rightarrow 0$. Uniform late-time parabolic Hölder bounds upgrade this to

$$\|u(t_j) - U\|_{L^\infty} \rightarrow 0. \quad (23)$$

Indeed, otherwise a point with deviation at least ε , together with a uniform spatial modulus of continuity, would produce an interval of fixed positive length on which the deviation is at least $\varepsilon/2$, contradicting the L^2 convergence.

Finally, $\chi_0 < \chi_{\text{lin}}$ supplies a spectral gap on the mass-zero subspace. The nonlinear remainder is quadratic in a fractional domain embedded in C^1 . The analytic semigroup estimate and variation of constants give, for sufficiently small data,

$$\|(u - U, v - v_*)(t)\|_{C^1} \leq C e^{-\rho(t-t_j)} \|u(t_j) - U\|_{L^\infty}.$$

Parabolic smoothing during a fixed positive time connects the L^∞ -small slice (23) to that fractional domain. This proves (5); the supremum convergence follows immediately.

3.3 Proof of the signal-energy branch

Assume $\gamma = 1$. Then $v_* = \nu U/\mu$, and integration of the elliptic equation gives

$$\int_0^1 (v - v_*) dx = 0.$$

Define

$$\mathcal{E}_v(t) = \int_0^1 [v_x^2 + \mu(v - v_*)^2] dx. \quad (24)$$

Differentiating the elliptic equation in time, pairing with $v - v_*$, and using the population equation gives the exact identity

$$\begin{aligned} \mathcal{E}_v'(t) = & -2 \int_0^1 v_{xx}^2 - 2\mu \int_0^1 v_x^2 \\ & + 2\nu\chi_0 \int_0^1 \frac{u^m v_x^2}{(1+v)^\beta}. \end{aligned} \quad (25)$$

On the eventual box,

$$\mathcal{E}_v'(t) \leq -2 \int_0^1 v_{xx}^2 - 2C_S \int_0^1 v_x^2, \quad C_S = \mu - \frac{\nu\chi_0 \bar{u}^m}{(1+\underline{v})^\beta}.$$

The last entry of (15) makes $C_S > 0$. Since the signal perturbation has zero mean, Poincaré's inequality gives

$$\mathcal{E}_v(t) \leq (\mu + 1) \int_0^1 v_x^2.$$

Hence $\mathcal{E}_v$ decays exponentially. Integration of (25) also supplies arbitrarily late slices with small $\|v_{xx}\|_{L^2}$. Because

$$u - U = \frac{\mu(v - v_*) - v_{xx}}{\nu},$$

such slices are L^2 -close to U ; smoothing and the one-dimensional embedding give the same L^∞ basin entry as in (23). The linear spectral bootstrap then proves (5). The first two entries in (15) are the smallness bounds used to produce the eventual box and its persistence estimates.

3.4 Why the same proof reaches $m = 2$

The entropy and signal-energy arguments above use only $m \geq 1$ after the eventual box has been obtained. The restriction $m < 2$ enters earlier, in the L^P recurrence that produces (11). At $m = 2$, the recurrence is critical. Condition (12) makes the coefficient of the restarted moment positive for an exponent $P > \max\{2, \gamma\}$. The endpoint Agmon estimate then yields (11), after which the same calculations (17)–(3.3) apply without change. Thus $m = 2$ is an endpoint of this a priori estimate, not a dynamical bifurcation value.

4 Every $m > 1$: a separate small-sensitivity route

For $m > 1$, define

$$G(t) = \int_0^1 u(t, x)^{-2m} u_x(t, x)^2 dx. \quad (26)$$

The negative-power entropy identity and the sharp saturation inequality

$$\frac{v}{(1+v)^\beta} \leq \sigma_\beta$$

give a parameter-only differential inequality. Indeed, the elliptic log-gradient bound

$$|v_x| \leq \sqrt{\mu} v$$

implies

$$\int_0^1 \frac{v_x^2}{(1+v)^{2\beta}} \leq \mu \sigma_\beta^2.$$

The half-Young estimate (18) therefore yields

$$\mathcal{E}'_u(t) \leq -\frac{c_m}{2} \left(G(t) - \chi_0^2 \mu \sigma_\beta^2 \right). \quad (27)$$

It follows that, for every $T > 0$ and $\varepsilon > 0$, there is $t \geq T$ such that

$$G(t) < \chi_0^2 \mu \sigma_\beta^2 + \varepsilon. \quad (28)$$

Otherwise (27) would force $\mathcal{E}_u$ to decrease at a fixed negative rate for all $t \geq T$, contradicting $\mathcal{E}_u(t) \geq 0$.

Lemma 1 (negative-power oscillation). *For every positive C^1 profile u on $[0, 1]$,*

$$\left| u(x)^{1-m} - u(z)^{1-m} \right| \leq (m-1) \sqrt{G} \quad (x, z \in [0, 1]). \quad (29)$$

Proof. The fundamental theorem of calculus and Cauchy–Schwarz give

$$\begin{aligned} \left| u(x)^{1-m} - u(z)^{1-m} \right| &\leq (m-1) \int_0^1 u^{-m} |u_x| \\ &\leq (m-1) \left(\int_0^1 u^{-2m} u_x^2 \right)^{1/2}. \end{aligned}$$

□

For $0 < \delta < U$, set

$$r_m(U, \delta) = \min \left\{ U^{1-m} - (U + \delta)^{1-m}, \frac{(U - \delta)^{1-m} - U^{1-m}}{2} \right\}. \quad (30)$$

Both entries are positive because $1 - m < 0$. Continuity and the mass constraint provide a point z with $u(z) = U$. Hence (29) implies

$$(m - 1)\sqrt{G} < r_m(U, \delta) \implies \|u - U\|_{L^\infty} < \delta. \quad (31)$$

To verify (31), note that $s \mapsto s^{1-m}$ is strictly decreasing. A point with $u \geq U + \delta$ or $u \leq U - \delta$ would force the oscillation from U^{1-m} to be at least one of the two quantities used in (30).

Theorem 2 (all- $m > 1$ small-sensitivity branch). *Assume $m > 1$, $\chi_0 < \chi_{\text{lin}}$, and let $\delta \in (0, U)$ be any L^∞ radius contained in the local spectral basin of the equilibrium. If*

$$(m - 1)\chi_0\sqrt{\mu}\sigma_\beta < r_m(U, \delta), \quad (32)$$

then the equilibrium is globally attracting among positive bounded mass- U solutions.

Proof. Choose $\varepsilon > 0$ so small that

$$(m - 1)\sqrt{\chi_0^2\mu\sigma_\beta^2 + \varepsilon} < r_m(U, \delta).$$

At the good time supplied by (28), the oscillation estimate and (31) put the solution inside the local L^∞ -basin. Restarting the autonomous equation at that time and using the linearly stable semigroup bootstrap proves (5). $\square$

Equivalently, the explicit threshold associated with δ is

$$\chi_0 < \frac{r_m(U, \delta)}{(m - 1)\sqrt{\mu}\sigma_\beta}.$$

Linear stability guarantees the existence of at least one positive local basin radius $\delta < U$.

5 Steady-state bifurcation

5.1 Integrated steady equation

Specialize to

$$\beta = \gamma = \mu = \nu = 1.$$

At a positive steady state, integration of the zero-flux equation gives

$$u_x - \chi \frac{u^m}{1 + v} v_x = 0.$$

If $H'_m(u) = u^{-m}$, then

$$H_m(u) = \begin{cases} \log u, & m = 1, \\ \frac{u^{1-m}}{1-m}, & m \neq 1, \end{cases}$$

and every positive steady state satisfies

$$H_m(u) - \chi \log(1 + v) = C, \quad -v_{xx} + v = u, \quad \int_0^1 u = U. \quad (33)$$

5.2 Weakly nonlinear direction coefficient

Let

$$e_1(x) = \cos(\pi x), \quad e_2(x) = \cos(2\pi x),$$

and set

$$r_1 = \frac{1}{1 + \pi^2}, \quad r_2 = \frac{1}{1 + 4\pi^2}, \quad b = 1 + U.$$

Seek a mass-preserving branch

$$u = U + \varepsilon e_1 + \varepsilon^2 a_2 e_2 + O(\varepsilon^3), \quad (34)$$

$$v = U + \varepsilon r_1 e_1 + \varepsilon^2 a_2 r_2 e_2 + O(\varepsilon^3), \quad (35)$$

$$\chi = \chi_{\text{lin}} + \varepsilon^2 \chi_2 + O(\varepsilon^3). \quad (36)$$

Substitution into (33) gives, successively,

$$\chi_{\text{lin}} = \frac{b}{r_1 U^m}, \quad (37)$$

$$a_2 = \frac{m/U - r_1/b}{4(1 - r_2/r_1)}, \quad (38)$$

$$\chi_2 = \chi_{\text{lin}} S(m, U), \quad (39)$$

$$S(m, U) = \frac{a_2}{2} \left(\frac{r_2}{b} - \frac{m}{U} \right) + \frac{m(m+1)}{8U^2} - \frac{r_1^2}{4b^2}. \quad (40)$$

The order- ε equation yields (37); the e_2 -projection at order ε^2 yields (38); and the e_1 -solvability condition at order ε^3 yields (39)–(40).

Thus $S > 0$ gives a forward branch and $S < 0$ gives a backward branch. The roots of the explicit function S are approximately

U	$m_c(U)$
0.5	2.866754572143
1	2.936658202934
2	3.005668668328

within the normalized family. These values describe only the direction of the local steady branch; they do not constitute a global stability diagram.

6 An exact subcritical steady state at $m = 3$

6.1 Banach-space branch construction

Fix

$$m = 3, \quad \beta = \gamma = \mu = \nu = U = 1.$$

Let $\mathcal{A}_2$ be the real even Wiener space

$$f(x) = \sum_{n \in \mathbb{Z}} f_n e^{in\pi x}, \quad f_{-n} = f_n \in \mathbb{R}, \quad \|f\|_{\mathcal{A}_2} = \sum_{n \in \mathbb{Z}} (1 + |n|)^2 |f_n| < \infty.$$

It is a Banach algebra and embeds continuously into C^2 . Let $R = (-\partial_{xx} + 1)^{-1}$, so

$$R e_n = \frac{1}{1 + n^2 \pi^2} e_n.$$

Restrict to the mean-zero subspace and write

$$h = e_1 + z, \quad z \perp e_1.$$

For amplitude a , set

$$u = 1 + ah, \quad v = 1 + aRh.$$

After dividing the steady flux equation by a , define

$$\mathcal{F}(a, z, \chi) = h - \chi \partial_x^{-1} \left[\frac{(1 + ah)^3}{2 + aRh} \partial_x Rh \right]. \quad (41)$$

The inverse derivative is taken on odd mean-zero Fourier series. Near $(a, z, \chi) = (0, 0, 2(1 + \pi^2))$, the denominator is invertible in the Banach algebra, so $\mathcal{F}$ is C^3 .

At the base point, the derivative in the complement direction is

$$L = I - (1 + \pi^2)R.$$

On the n -th cosine mode,

$$Le_n = \frac{(n^2 - 1)\pi^2}{1 + n^2\pi^2} e_n. \quad (42)$$

Its kernel on the mean-zero space is precisely the first cosine mode, which has been removed from z . For $n \geq 2$, the reciprocal multiplier is

$$\frac{1 + n^2\pi^2}{(n^2 - 1)\pi^2},$$

uniformly bounded in n , so L has a bounded inverse on the complement. The derivative in the sensitivity direction is

$$-\frac{1}{2(1 + \pi^2)} e_1,$$

which spans the missing first-mode line. Hence

$$D_{(z, \chi)} \mathcal{F}: \{e_1\}^\perp \times \mathbb{R} \longrightarrow \{\text{mean-zero even profiles}\}$$

is a bounded isomorphism.

The Banach implicit-function theorem now gives $a_0 > 0$ and a C^3 branch

$$a \longmapsto (z(a), \chi(a)), \quad z(0) = 0, \quad \chi(0) = 2(1 + \pi^2),$$

for $|a| < a_0$, satisfying $\mathcal{F} = 0$.

6.2 Exact branch direction

Multiplying (41) by the signal denominator and differentiating gives the polynomial flux identity

$$(2 + aRh) \partial_x h = \chi(a)(1 + ah)^3 \partial_x Rh. \quad (43)$$

Projecting the first derivative of (43) onto e_1 gives $\chi'(0) = 0$. The mode e_2 determines $z'(0)$, equivalently the coefficient a_2 in (38). Projecting the second derivative onto e_1 , or substituting $m = 3, U = 1$ into (38)–(40), gives

$$\chi''(0) = -\frac{6\pi^4 + 37\pi^2 + 25}{24\pi^2(\pi^2 + 1)} < 0. \quad (44)$$

Every factor in the denominator is positive and the numerator is positive, so the sign is strict.

By continuity of χ'' , after reducing a_0 if necessary, $\chi'(a) < 0$ for $0 < a < a_0$. Since $\chi(0) > 0$, there is $a_1 \in (0, a_0)$ such that

$$0 < \chi(a) < 2(1 + \pi^2) \quad (0 < a < a_1). \quad (45)$$

6.3 Positivity and the dynamical obstruction

The embedding $\mathcal{A}_2 \hookrightarrow C^2$ and $z(a) \rightarrow 0$ show that, for small $a > 0$,

$$u_a = 1 + a(e_1 + z(a)) > 0, \quad v_a = 1 + aR(e_1 + z(a)) > 0.$$

The cosine representation gives the Neumann boundary conditions, the missing zero Fourier coefficient gives

$$\int_0^1 u_a dx = 1,$$

and the normalized first coefficient makes u_a nonconstant. Differentiating (43) recovers

$$\begin{aligned} 0 &= (u_a)_{xx} - \chi(a) \partial_x \left(\frac{u_a^3 (v_a)_x}{1 + v_a} \right), \\ 0 &= (v_a)_{xx} - v_a + u_a. \end{aligned}$$

Theorem 3 (subcritical nonconstant steady state). *There exist $0 < \chi < 2(1 + \pi^2)$ and positive, nonconstant C^2 Neumann profiles u, v , of mass one, satisfying the $m = 3$ steady system.*

Regard these profiles as constant-in-time functions. They are a positive, bounded, global classical orbit. If the constant equilibrium $(1, 1)$ attracted every such orbit, then the time-independent function u would converge uniformly to 1, forcing $u \equiv 1$, contrary to nonconstancy.

Theorem 4 (failure of global attraction below the linear threshold). *For $m = 3$, $\beta = \gamma = \mu = \nu = U = 1$, there is a sensitivity $0 < \chi < \chi_{\text{lin}}$ for which the constant equilibrium is linearly stable but not globally attracting on the mass-one hyperplane.*

7 Resolved region and remaining gap

The proved picture is:

1. for $1 \leq m < 2$, the explicit entropy or signal-energy conditions (E) – (S) imply global attraction;
2. $m = 2$ is included under (12);
3. for every $m > 1$, (32) gives a separate small-sensitivity stability region;
4. at $m = 3, U = 1$, a nonconstant positive steady state exists strictly below the linear threshold, so linear stability does not imply global attraction.

No necessary-and-sufficient global stability diagram is claimed in the region between the sufficient thresholds and the backward steady branch. In particular, $m < m_c(U)$ only means that the first local branch points forward; it does not prove global attraction all the way to χ_{lin} . Conversely, $m > m_c(U)$ predicts a local backward branch, but the analytic branch construction above was carried out for the fixed $m = 3, U = 1$ tuple.